1. **What AI tools did your team use (if any)?**

Our team mainly used GPT-5.6 Terra to help write, revise, and debug our Python code. To review this code that GPT wrote, we also used GitHub Copilot in order to identify areas that need improvement. Copilot helped us put together a more detailed final prompt that included all of the code requirements. Using both the tools of GPT and Copilot, we were able to successfully and efficiently improve our final prompt, which helped us with our code.

2. **Did the AI tools your team used solve the problem correctly on the first try? If not, how many tries did it take?**

No, the AI tools did not solve the problem correctly on the first try. It took around 25 attempts (prompts) before we were able to successfully get a solution that met all assignment requirements. Each attempt helped us identify something new that we either needed to improve or change. The assignment made sure we had to be very specific about the rules, since the program had many different variations for the truffle boxes.

3. **What prompt(s) produced the correct solution?**

The final prompt included the GitHub assignment, the original code, and all the updated requirements for the truffle box assignment. The prompt explained all the restrictions with the shapes, fillings, toppings, and chocolate types. There was also the requirement that each box contain either 2 or 3 white chocolate truffles. We created the final prompt below using GPT, checked by Copilot, to address all the rules and fix issues from earlier attempts.

Final Prompt:

Using https://github.com/tamu-edu-students/engr-102-lab-10-team and the following feedback:

```
I changed my mind... I don't want toffee and coconut fillings in the same box
And also... I don't want both white and dark chocolate drizzle toppings in the same box
I bought new boxes... so now I don't want round and heart shapped truffles in the same box
My friends changed their minds... They want at least two and no more than three white chocolate truffles in the same box
Must have either 2 or 3 white chocolate truffles in the same box!
```

Revise my Python code to solve this truffle-box assignment. Keep the existing `make_boxes(chocolates, time_limit=4)` interface and make the behavior match these requirements:

* Each box contains exactly four truffles, and every input truffle ID is used exactly once.
* If the number of truffles is not divisible by four, return an empty list.
* Two truffles with identical four attributes cannot be in the same box.
* These pairs cannot be in the same box: rectangle and square shapes; round and heart shapes; caramel and vanilla fillings; toffee and coconut fillings; nuts and sprinkles toppings; white chocolate drizzle and dark chocolate drizzle.
* Each box must contain 0, 2, or 3 white-chocolate truffles. **Do not add a dark-chocolate count restriction.**

* Normalize inputs by lowercasing and collapsing whitespace. Treat “nothing,” “no topping,” and an empty topping as “none.” Recognize descriptions such as “heart shaped,” “round shaped,” “vanilla creme,” “strawberry creme,” “coconut creme,” “caramel creme,” and “toffee creme” as their corresponding standard values.
* Search for a valid arrangement for up to `time_limit` seconds. If one is found, return it. If not, return the arrangement found that breaks the fewest rules rather than raising an error. If no complete arrangement is found, return an empty list.
Include a `check_boxes(chocolates, boxes)` function that returns `True` only if all IDs are used exactly once and every box meets the rules above. Keep the implementation readable, explain the key changes, and include a small test using sample input.

4. **Did the results surprise you? Why (not)?**

The results did not surprise me too much, because generally AI is helpful with coding and programming. However, I was surprised by how many attempts it took to actually get the code right. Even though AI works effectively, it sometimes overlooked requirements that we had displayed, or introduced new problems to fix. This shows me that AI is helpful, but still needs to be looked over for new problems that arise.

5. **Do you plan to use AI tools again (future courses, your career, personal life, etc)? Why (not)?**

Yes, we plan to continue to use AI in the future, but also making sure that it is used well is just as important. AI can save time, explain difficult concepts, and help me work out problems that I might have struggled with. As engineering students, we think AI is extremely useful and helpful when it comes to problem solving. However, we also want to make sure that we know the concepts instead of relying purely on AI to do them.

6. **Briefly explain how AI tools can (or can't) be helpful for debugging code.**

AI tools can be very helpful when debugging because they can identify errors in code with ease compared to a human. They review larger sections of code and point out problems that might be difficult to find on your own. This does not mean it always finds the correct issue, though. Sometimes AI can introduce additional errors into your programming. It is important to check the suggested changes and understand why they work.

7. **Briefly describe what you found to be the most difficult part of this assignment.**

The most difficult part of this assignment was all of the rules the program had to follow when creating the different truffle boxes. There were many possible variations in all of the shapes, fillings, toppings, and flavors that had set rules. It was challenging to make sure every truffle was used correctly and exactly once while also following the white chocolate requirements. When inputting this all into AI it was hard to be so specific so that the AI knew what I wanted it to do.